

Dr Benjamin A. D. Williamson

MSci MPhil PhD



Senior Researcher
Department of Materials Science and Engineering
Norwegian University of Science and Technology (NTNU)
7491 Trondheim
Norway

Email ✉ : benjamin.williamson@ntnu.no
Website/Github 📄 : badw.github.io
Google Scholar 📖 : scholar.google.com/citations?user=c34AgtMAAAAJ
LinkedIn in : [benjamin-williamson-8b7b932a4](https://www.linkedin.com/in/benjamin-williamson-8b7b932a4)
Nationality 🇬🇧 : British

🔧 Employment

Norwegian University of Science and Technology (NTNU) (2019 – Present)

Trondheim, Norway

Senior Researcher in Computational Materials Design

Themes: Battery materials, ferroelectrics/multiferroics, solar cells, novel inorganic/organic functional materials
Functional Materials and Materials Chemistry Research Group (FACET)

Advisors: Prof. Tor Grande & Prof. Sverre M. Selbach

University College London (2018 – 2019)

London, United Kingdom

Research Associate in Computational Materials Design

Scanlon Materials Theory Group

Advisor: Prof. David O. Scanlon

🎓 University Education

University College London (2014 – 2018)

London, United Kingdom

MPhil/PhD in Computational Inorganic Materials Chemistry – *no corrections*

Thesis title : *Understanding the Electronic and Thermodynamic Properties of Wide Band Gap Materials*

Supervisor: Prof. David O. Scanlon; Secondary Supervisor: Prof. Claire J. Carmalt

University College London (2010 – 2014)

London, United Kingdom

MSci in Chemistry – 1st Class Honours

MSci dissertation title: *Computational Design of Next-Generation p-Type Semiconductors*

Supervisor: Prof. David O. Scanlon

💰 Funding Grants

Tuneable Earth Abundant and Solid-State Electrolytes for Li-ion Batteries (TEASE)

Research Council of Norway

8,000,000 NOK

The TEASE project aims to predict new Earth-abundant solid state electrolytes for safer energy dense Li-ion batteries using the latest computational simulation methods.

Peer Reviewed Publications

(In reverse chronological order)

As of September 2026:

- Total Citations: 2010

42. *Electronic Structure and Resonant Circular Dichroism of LaSrMnO from Soft X-ray Angle-Resolved Photoemission*
Ø. Finnseth, D. Brzozowski, A. C. Mathisen, S. S. Brinkman, X. L. Tan, F. Gohler, B. A. D. Williamson, K. Eggestad, M.-J. Huang, J. Buck, M. Hoesch, K. Rossnagel, S. M. Selbach, H. Bentmann, I. Hallsteinsen
Phys. Rev. B, 114, 185122, (2026)
DOI: <https://doi.org/10.1103/59pf-43kx>
41. *Local indirect magnetoelectric coupling at twin walls in CaMnO₃*
I.C. Skogvoll, **B.A.D. Williamson**, S.M. Selbach, *Phys. Rev. B* 113 224102 (2026)
<https://doi.org/10.1103/6y99-279q>
40. *Domain Walls and Defects in Ferroelectric Inorganic Halide Perovskites CsGeX₃ (X = Cl, Br, I)*
K. Eggestad, **B.A.D. Williamson**, S.M. Selbach, *Phys. Rev. Mater.* 10 064403 (2026):
<https://doi.org/10.1103/bf5v-24b2>
39. *Finite-temperature ferroelectric phase transitions from machine-learned force fields*
K. Eggestad, I. C. Skogvoll, Ø. Gullbrekken, **B.A.D. Williamson**, and S. M. Selbach, *Phys. Rev. Mater.*, 10 034409, (2026)
DOI: <https://doi.org/10.1103/nkc5-rtids>
38. *Spatial distribution of crystalline impurities in degraded high voltage XLPE cable screens using synchrotron wide-angle X-ray scattering*
S. B. Hårberg, O. G. Grendal, **B.A.D. Williamson**, H. Strand, M. Burghammer, S. Hvidsten, and M.-A. Einarsrud, *Journal of Physics: Energy*, 8 015016, (2026)
DOI: <https://iopscience.iop.org/article/10.1088/2515-7655/ae34e8>
37. *Quantifying Emptiness: On the Size of A-Site Vacancies in Tetragonal Tungsten Bronzes*
C. R. Zeiger, K. Eggestad, J. Walker, M.-A. Einarsrud, **B.A.D. Williamson**, and T. Grande *Chemistry of Materials*, ASAP, (2025)
DOI: 10.1021/acs.chemmater.4c03438
36. *Phase Evolution and Thermodynamics of Cubic Li_{6.25}Al_{0.25}La₃Zr₂O₁₂ Studied by High-Temperature X-ray Diffraction*
Ø. Gullbrekken, K. Eggestad, M. Tsoutsouva, **B.A.D. Williamson**, D. Rettenwander, M.-A. Einarsrud, and S. M. Selbach *Inorganic Chemistry*, 64, 12, 5866-5865, (2025)
DOI: 10.1021/acs.inorgchem.4c03738
35. *Pseudo-Lithium Vacancies in Hydrogen Rich Li₃OCl*
B.A.D. Williamson, K. Eggestad, S. M. Selbach *J. Mater. Chem. A*, 13, 8816-8824, (2025)
DOI: 10.1039/D4TA08352K
34. *Crystal Structure, Cation Occupation, and Phase Transitions in Ba₄(Li_xNa_{1-x})₂Nb₁₀O₃₀ Tetragonal Tungsten Bronzes*
N. S. Løndal, **B.A.D. Williamson**, O.G. Grendal, J. Walker, M.-E. Einarsrud, and T. Grande, *Inorganic Chemistry*, Advance Article, (2024)
DOI: 10.1021/acs.inorgchem.4c04461
33. *Mobile Intrinsic Point Defects for Conductive Neutral Domain Walls in LiNbO₃*
K. Eggestad, **B.A.D. Williamson**, D. Meier, and S. M. Selbach, *J. Mater. Chem. C*, 12, 17099-17107, (2024)
DOI: 10.1039/D4TC02856B
32. *Doping Implications of Li-ion Solid State Electrolyte Li₇La₃Zr₂O₁₂*
K. Eggestad, S. M. Selbach, and **B.A.D. Williamson**, *J. Mater. Chem. A*, 12, 15666-15675, (2024)
DOI: 10.1039/D4TA01487A
Work selected as one of the Journal of Materials Chemistry A HOT Papers 2024
31. *On the solubility of Bi in tetragonal tungsten bronzes*
C. R. Zeiger; **B.A.D. Williamson**; J. Walker, M.-A. Einarsrud, T. Grande. *J. Euro. Ceram. Soc.*, 44, 6, 3723-3733 (2024)
DOI: 10.1016/j.jeurceramsoc.2024.01.005

30. *The effect of cation size on structure and properties of Ba-based tetragonal tungsten bronzes $Ba_4M_2Nb_{10}O_{30}$ ($M = Na, K$ or Rb) and $Ba_4M_2Nb_8Ti_2O_{30}$ ($M = Ca$ or Sr)*
N. S. Løndal; **B.A.D. Williamson**; J. Walker, M.-A. Einarsrud, T. Grande. *Phys. Chem. Chem. Phys.* , 26, 3350-3366 (2024)
DOI: 10.1039/D3CP05666J
29. *Oxygen absorption in nanocrystalline $h\text{-}RMnO_3$ ($R=Y, Ho, Dy$) and the effect of Ti donor doping*
F. Danmo; **B.A.D. Williamson**; D. Småbråten, N. Gaukås, E.R. Østli, T. Grande; J. Glaum; S. Selbach; *Chem. Mater.* , 34, 15,5764-5776 (2023)
DOI: 10.1021/acs.chemmater.3c00189
28. *Multi-Phase Sputtered TiO_2 -Induced Current–Voltage Distortion in Sb_2Se_3 Solar Cells*
C.H. Don, T.P. Shalvey, M.J. Smiles, L. Thomas, L.J. Phillips, T.D.C. Hobson, H. Finch, L.A.H. Jones, J.E.N. Swallow, N. Fleck, C. Markwell, P.K. Thakur, T.-L. Lee, D. Biswas, L. Bowen, **B.A.D. Williamson**, D.O. Scanlon, V.R. Dhanak, K. Durose, T.D. Veal, J.D. Major; *Adv. Mater. Int.* , 2300238 (2023)
DOI: 10.1002/admi.202300238
27. *High-throughput calculations of charged point defect properties with semi-local density functional theory performance benchmarks for materials screening applications*
D. Broberg, K. Bystrom, S. Srivastava, D. Dahliah, **B.A.D. Williamson**, L. Weston, D. O. Scanlon, G.-M. Rignanesse, S. Dwaraknath, J. Varley, K. A. Persson, M. Asta, G. Hautier; *npj Comp. Mater.* , 9, 1, 72 (2023)
DOI: 10.17188/mpcontribs/1907859
26. *Tuning the Thermoelectric Performance of $CaMnO_3$ -based ceramics by controlled exsolution and microstructuring*
N. Kanas, **B.A.D. Williamson**, F. Steinbach, R. Hinterding, M.-A. Einarsrud, S.M. Selbach, A. Feldhoff and K. Wiik; *ACS Appl. Energy Mater.* , 5, 10, 12396–12407 (2022)
DOI: 10.1021/acsaem.2c02012
25. *Computational Prediction and Experimental Realisation of Earth Abundant Transparent Conducting Oxide Gated $ZnSb_2O_6$*
A. J. Jackson, B. J. Parrett, J. Willis, A. M. Ganose, W. W. W. Leung, **B.A.D. Williamson**, Y. Liu, T. K. Kim, M. Hoesch, L. Ishibe-Veiga, R. Kalra, J. Neu, C. A. Schmuttenmaer, T.-L. Lee, A. Regoutz, T. D. Veal, R. G. Palgrave, R. Perry and D. O. Scanlon; *ACS Energy Lett.* 2022, 7, XXX, 3807–3816, (2022)
DOI: 10.1021/acsenergylett.2c01961
24. *Mesophase transition in $[(C_2H_5)N][FeBrCl_3]$ and $[(CH_3)_4N][FeBrCl_3]$ ferroic plastic crystals*
J. Walker, K. P. Marshall, **B.A.D. Williamson**, J.S. Beceiro, N. S. Løndal, S. M. Selbach, S. C. Garcia, D. Chernyshov, M.-A. Einarsrud; *Chem. Mater.*, 34, 6, 2585-2598, (2022)
DOI: 10.1021/acs.chemmater.1c03778
23. *Investigation of factors affecting the stability of compounds formed by isovalent substitution in layered oxychalcogenides, leading to identification of $Ba_3Sc_2O_5Cu_2Se_2$, $Ba_3Y_2O_5Cu_2S_2$, $Ba_3Sc_2O_5Ag_2Se_2$ and $Ba_3In_2O_5Ag_2Se_2$*
G. J. Limburn, D. W. Davies, N. Langridge, Z. Malik, **B. A. D. Williamson**, D. O. Scanlon, and G. Hyett; *J. Mater. Chem. C*, 10, 3784-3795, (2022)
DOI: 10.1039/D1TC05051F
22. *Ligand Field-Induced Exotic Dopant for Infrared Transparent Electrode: W in Rutile SnO_2*
M. Fukumoto, Y. Hirose, **B.A.D. Williamson**, S. Nakao, K. Kimura, K. Hayashi, Y. Sugisawa, D. Sekiba, D.O. Scanlon and T. Hasegawa; *Adv. Func. Mater.*, 2110832, (2021)
DOI: 10.1002/adfm.202110832
21. *$BaBi_2O_6$: A Promising n-Type Thermoelectric Oxide with the $PbSb_2O_6$ Crystal Structure*
K.B. Spooner, A.M. Ganose, W.W.W. Leung, J. Buckeridge, **B.A.D. Williamson**, R.G. Palgrave, and D.O. Scanlon, *Chem. Mater.*, <https://pubs.acs.org/doi/10.1021/acs.jpcc.0c11592>, 33, 7441, (2021)
DOI: 10.1021/acs.chemmater.1c02164
20. *Experimental and theoretical study of the electronic structures of lanthanide indium perovskites $LnInO_3$*
P. Hartley, K.H.L. Zhang, M.V. Hohmann, L.F.J. Piper, D.J. Morgan, R.G. Edgell, D.O. Scanlon, **B.A.D. Williamson**, and A. Regoutz; *J. Phys. Chem. C*, 125, 6387, (2021)
DOI: 10.1021/acs.jpcc.0c11592

19. *Photocatalytic, Structural and Optical Properties of Mixed Anion Solid Solutions $Ba_3Sc_{2-x}In_xCu_2S_2$ and $Ba_3In_2O_5Cu_2S_{2-y}Se_y$*
G. Limburn, M. Stephens, **B.A.D. Williamson**, A. Iborra-Torres, D.O. Scanlon and G. Hyett; *J. Mater. Chem. A*, 8, 19887-19897 (2020)
DOI: 10.1039/D0TA06629J
18. *Computationally Driven Discovery of Layered Quinary Oxychalcogenides: Potential p-Type Transparent Conductors?*
B.A.D. Williamson, G.J. Limburn, G. Hyett, G. W. Watson and D.O. Scanlon; *Matter; Cell Press*, 3,3, 759-781 (2020)
DOI: 10.1016/j.matt.2020.05.020
Article Preview by A. Walsh and J-S. Park: The Holey Grail of Transparent Electronics
DOI: 10.1016/j.matt.2020.08.008
17. *Computational Prediction of the Thermoelectric Performance of $LaZnOPn$ ($Pn = P, As$)*
M. Einhorn, **B.A.D. Williamson** and D.O. Scanlon; *J. Mater. Chem. A*, 8, 7914-7924 (2020)
DOI: 10.1039/D0TA00690D
16. *Resonant Ta Doping for Enhanced Mobility in Transparent Conducting SnO_2*
B.A.D. Williamson, T.J. Featherstone, S. Sathasivam, J.E.N Swallow, H. Shiel, L.A.H. Jones, M.J. Smiles, A. Regoutz, T-L. Lee, X. Xia, C. Blackman, P.K. Thakur, C.J. Carmalt, I.P. Parkin, T.D. Veal and D.O. Scanlon; *Chem. Mater.*, 32, 5, 1964-1973 (2020)
DOI: 10.1021/acs.chemmater.9b04845
Work featured on the front cover
15. *Enhanced Photocatalytic and Antibacterial Ability of Cu-doped Anatase TiO_2 Thin Films: Theory and Experiment*
A.M. Alotaibi, **B.A.D. Williamson**, S. Sathasivam, A. Kafizas, M. Alqahtani, C. Sotelo-Vazquez, J. Buckeridge, J. Wu, S.P. Nair, D.O. Scanlon and I.P. Parkin; *ACS Appl. Mater. Interfaces* just accepted manuscripts (2020)
DOI: 10.1021/acsami.9b22056
14. *Resonant Doping for High Mobility Transparent Conductors: The Case of Mo-doped In_2O_3*
J.E.N. Swallow, **B.A.D. Williamson**, S. Sathasivam, M. Birkett, T.J. Featherstone, P.A.E. Murgatroyd, H.J. Edwards, Z.W. Lebens-Higgins, D.A. Duncan, M. Farnworth, P. Warren, N. Peng, T-L. Lee, L.F.J. Piper, A. Regoutz, C.J. Carmalt, I.P. Parkin, V.R. Dhanak, D.O. Scanlon and T.D. Veal; *Mater. Horiz.*, 7, 236-243 (2020)
DOI: 10.1039/C9MH01014A
13. *Dispelling the Myth of Passivated Codoping in TiO_2*
B.A.D. Williamson, J. Buckeridge, N.P. Chadwick, S. Sathasivam, C.J. Carmalt, I.P. Parkin and D.O. Scanlon; *Chem. Mater.*, 31 (7), 2577-2589 (2019)
DOI: 10.1021/acs.chemmater.9b00257
12. *Origin of High-Efficiency Photoelectrochemical Water Splitting on Hematite/Functional Nanohybrid Metal Oxide Overlayer Photoanode after a Low Temperature Inert Gas Annealing Treatment*
S. Ho-Kimura, **B.A.D. Williamson**, S. Sathasivam, S.J.A. Moniz, G. He, W. Luo, D.O. Scanlon, J. Tang, I.P. Parkin; *ACS Omega*, 4 (1), 1449-1459 (2019)
DOI: 10.1021/acsomega.8b02444
11. *Phosphorus Doped SnO_2 Thin Films for Transparent Conducting Oxide Applications: Synthesis, Optoelectronic Properties and Computational Models*
M.J. Powell, **B.A.D. Williamson**, S-Y. Baek, J. Manzi, D. Potter, D.O. Scanlon and C.J. Carmalt; *Chem. Sci.*, 9 (41), 7968-7980 (2018)
DOI: 10.1039/C8SC02152J
10. *Enhanced Electrical Properties of Antimony Doped Tin Oxide Thin Films Deposited via Aerosol Assisted Chemical Vapour Deposition*
S. Ponja, **B.A.D. Williamson**, S. Sathasivam, D.O. Scanlon, I.P. Parkin, C.J. Carmalt; *J. Mater. Chem. C*, 6, 7257-7266 (2018)
DOI: 10.1039/C8TC01929K
9. *A Novel Laboratory-based Hard X-ray Photoelectron Spectroscopy System*
A. Regoutz, M. Mascheck, T. Wiell, S.K. Eriksson, C. Liljenberg, K. Tetzner, **B.A.D. Williamson**, D. O. Scanlon and P. Palmgren; *Rev. Sci. Inst.*, 89 (7), 073105 (2018)
DOI: 10.1063/1.5039829

8. *Chemical Vapor Deposition of Photocatalytically Active Pure Brookite TiO₂ Thin Films*
A.M. Alotaibi, S. Sathasivam, **B.A.D. Williamson**, A. Kafizas, C. Sotelo-Vazquez, A. Taylor, D.O. Scanlon, and I.P. Parkin; *Chem. Mater.*, 30 (4), 1353-1361 (2018)
DOI: 10.1021/acs.chemmater.7b04944
7. *A Deeper Understanding of Interstitial Boron-Doped Anatase Thin Films as A Multifunctional Layer Through Theory and Experiment*
M. Quesada-Gonzalez, **B.A.D. Williamson**, C. Sotelo-Vazquez, A. Kafizas, N.D. Boscher, R. Quesada-Cabrera, D.O. Scanlon, C.J. Carmalt, I.P. Parkin; *J. Phys. Chem. C*, 122 (1), 714-726 (2018)
DOI: 10.1021/acs.jpcc.7b11142
6. *Self-Compensation in Transparent Conducting F-Doped SnO₂*
J.E.N. Swallow, **B.A.D. Williamson**, T.J. Whittles, M. Birkett, T.J. Featherstone, N. Peng, A. Abbott, M. Farnworth, K.J. Cheetham, P. Warren, D.O. Scanlon, V.R. Dhanak, T.D. Veal; *Adv. Funct. Mater.*, 1701900 (2017)
DOI: 10.1002/adfm.201701900
5. *Chemical Vapor Deposition Synthesis and Optical Properties of Nb₂O₅ Thin Films with Hybrid Functional Theoretical Insight into Band Structure and Band Gaps*
S. Sathasivam, **B.A.D. Williamson**, S.A. Al Thabaiti, A.Y. Obaid, S.N. Basahel, M. Mokhtar, D.O. Scanlon, C.J. Carmalt, I.P. Parkin; *ACS Appl. Mater. Interfaces*, 9 (21), 18031-18038 (2017)
DOI: 10.1021/acsami.7b00907
4. *Computational and Experimental Study of Ta₂O₅ Thin Films*
S. Sathasivam, **B.A.D. Williamson**, A. Kafizas, S.A. Althabaiti, A.Y. Obaid, S.N. Basahel, D.O. Scanlon, C.J. Carmalt, I.P. Parkin; *J. Phys. Chem. C*, 121 (1), 202-210 (2017)
DOI: 10.1021/acs.jpcc.6b11073
3. *Transparent Conducting n-type ZnO:Sc – Synthesis, Optoelectronic Properties and Theoretical Insight*
S.C. Dixon, S. Sathasivam, **B.A.D. Williamson**, D.O. Scanlon, C.J. Carmalt, I.P. Parkin; *J. Mater. Chem. C*, 5, 7585-7597 (2017)
DOI: 10.1039/C7TC02389H
2. *Engineering Valence Band Dispersion for High Mobility p-Type Semiconductors*
B.A.D. Williamson, J. Buckeridge, J. Brown, S. Ansbro, R.G. Palgrave, D.O. Scanlon; *Chem. Mater.*, 29 (6), 2402-2413 (2017)
DOI: 10.1021/acs.chemmater.6b03306
Work featured as part of the front cover
1. *A Single-Source Precursor Approach to Solution Processed Indium Arsenide Thin Films*
P. Marchand, S. Sathasivam, **B.A.D. Williamson**, D. Pugh, S.M. Bawaked, S.N. Basahel, A.Y. Obaid, D.O. Scanlon, I.P. Parkin, C.J. Carmalt; *J. Mater. Chem. C*, 4, 6761-6768 (2016)
DOI: 10.1039/C6TC02293F

In Submission

4. *Machine-learning-guided exploration of domain walls in the hybrid improper ferroelectric Ca₃Ti₂O₇*
I. C. Skogvoll, E. Fransson, L. Ö. Westin, **B. A. D. Williamson**, N. C. Bristowe, S. M. Selbach, P. Erhart
arXiv, (2026) DOI: <https://doi.org/10.48550/arXiv.2609.35346>
3. *Stress-modulated ionic transport limits composite cathode kinetics in solid-state batteries*
H. Q. Nguyen, J. Hörmann, **B. A. D. Williamson**, M. Kanedal, J. Todt, F. Maletić, S. Erker, G. J. Redhammer, N. Schell, J. Keckes, A. Latz, T. Danner, and D. Rettenwander
chemrxiv, (2026) DOI: <https://doi.org/10.26434/chemrxiv.15002843/v>
2. *Resonant XX Doping for High-Mobility In₂O₃-Based Conductors with Enhanced Near-Infrared Transparency*
T. J. Featherstone, J. Willis, J.E.N. Swallow, **B.A.D. Williamson**, L.A.H. Jones, T.-Li. Lee, P.K. Thakur, G.W. Watson, T. Koida, D.O. Scanlon, and T.D. Veal; *in submission*
1. *Effect of pseudo-cubic (111)-oriented orthorhombic substrate facets on perovskite oxide thin film synthesis*
K. Kjærnes, T. Bolstad, D.M. Evans, E. Lysne, **B.A.D. Williamson**, D. Meier, S.M. Selbach, T. Tybell; *in submission*

Conference Presentations

1. **Invited:** B.A.D Williamson: *Understanding CO₂ Transport using ARCS: Automated Reactions for CO₂ Storage*, MATHIAS Days conference, Total Energies, Paris, France 2024 – Talk
2. **Contributed:** B.A.D Williamson: *Computational Analysis of a Promising Earth Abundant, Stable, Lithium Solid Electrolyte.*, MRS Spring, Seattle, USA 2024 – Talk
3. **Invited:** B.A.D Williamson: *Approaches to Overcoming the Challenges of Ab Initio Prediction of Functional Plastic Crystals.*, NMBU Advancing Functional Molecular Matter , NMBU, Ås, Norway 2024 – Talk
4. **Contributed:** B.A.D. Williamson *First Principles Insights into Phase Transitions, Disorder and Electronic Properties of Lead-Free Ba-based Tetragonal Tungsten Bronzes.*, IMF Ferroelectrics, Tel Aviv, Israel 2023 –Talk
5. **Contributed:** B.A.D. Williamson *Can we Utilise Phonons to Enhance Li-ion Diffusion?*, MRS Fall Meeting, Boston, USA 2022 –Talk
6. **Contributed:** B.A.D. Williamson *The Underlying Mechanisms of Ultrahigh Mobility in Bi₂O₂Se using Self-Consistent GW Method*, MRS Spring Meeting, Honolulu, USA 2022 –Talk
7. **Contributed:** B.A.D Williamson: *Computational discovery of an earth abundant, stable Li solid electrolyte*, FACET Sustainability Workshop, Trondheim, Norway 2021 –Talk
8. **Contributed:** B.A.D Williamson: *Computational discovery of an earth abundant, stable Li solid electrolyte*, FACET Sustainability Workshop, Trondheim, Norway 2021 –Talk
9. **Contributed:** B.A.D Williamson: *Computational Insights into Li diffusion in Solid State Electrolytes*, 7th Inorganic and Materials Norsk Kjemisk Selskap (NKS) Meeting, Værnes, Norway 2020 –Talk
10. **Contributed:** B.A.D Williamson: *Resonant Doping for High Mobility Transparent Conductors: The Case of Mo-doped In₂O₃*, 11th Petite Workshop on defects in energy materials; Sommarøy, Norway 2019 – Talk
11. **Contributed:** B.A.D Williamson: *Computationally Aided Discovery of Layered Quinary Oxychalcogenide p-type Transparent Conductor*, Workshop For Atomistic Modelling; Trondheim, Norway 2019 – Poster
12. **Contributed:** B.A.D Williamson: *Dispelling the Myth of Passivated Codoping in TiO₂*, MRS Fall Meeting; Boston US, 2018 – Talk
13. **Contributed:** B.A.D.Williamson: *Computationally Aided Discovery of Layered Quinary Oxychalcogenide p-type Transparent Conductors*, MRS Fall Meeting; Boston US, 2018 – Poster
Winner of the ICDD prize for materials characterisation
14. **Contributed:** B.A.D Williamson: *Doubled Conductivity in Transparent Conducting In₂O₃ Through Novel Dopant Design*, MMM Hub; Thomas Young Centre, London, UK, 2018 – Poster
15. **Invited:** B.A.D Williamson: *Doubled Conductivity in Transparent Conducting In₂O₃ Through Novel Dopant Design*, MCC 3rd Conference, Lincoln UK, 2018 – Talk
16. **Contributed:** B.A.D Williamson: *Doubled Conductivity in Transparent Conducting In₂O₃ Through Novel Dopant Design*, Gordon Research Conference; Defects In Semiconductors, Colby-Sawyer College, New Hampshire, US, 2018 – Poster
17. **Invited:** B.A.D Williamson: *Beyond Conventional Doping in SnO₂*, Thomas Young Centre, London UK, 2017 – Talk
18. **Contributed:** B.A.D Williamson: *Beyond Conventional Doping in SnO₂*, E-MRS; Spring Meeting, Strasbourg France, 2017 – Poster
19. **Contributed:** B.A.D Williamson: *Engineering Valence Band Dispersion For High-Mobility p-type Semiconductors*, E-MRS; Spring Meeting, Strasbourg France, 2017 – Talk
20. **Contributed:** B.A.D Williamson: *Engineering Valence Band Dispersion For High-Mobility p-type Semiconductors*, MRS Fall Meeting; Boston US, 2016 – Poster
21. **Contributed:** B.A.D Williamson: *Engineering Valence Band Dispersion For High-Mobility p-type Semiconductors*, SSCG Christmas Meeting; Canterbury UK, 2015 – Poster

Industrial Collaborations

2023 – Present Hydro (www.hydro.com)
2021 – Present: Equinor (www.equinor.com)
 Northern Lights (www.northernlightscs.com)
 Total Energies (www.totalenergies.com)
 Shell (www.shell.com)
2014 – 2019: Pilkingtons NSG Group (www.pilkington.com)

Teaching and Mentoring

2025 – Present: Main supervisor to 1 PhD student
2022 – Present: Co-supervisor to 3 PhD students
2019 – Present: Proposed and supervised 7 final year Masters research projects at NTNU
2021: Lecturer of Masters Course "TMT4245 FUNCTIONAL MATERIALS" at NTNU
2014 – 2019: Supervised 6 final year MSci research projects at UCL
2014 – 2019: Demonstrated in 1st year workshops (CHEM1004)
2018 – 2019: Tutor in 1st year inorganic chemistry courses at UCL (CHEM1101 and CHEM0013)

Computational Competance

UNIX/BASH/ZSH proficiency: high
Python proficiency: high
LaTEX proficiency: high
VASP proficiency: high
Adobe Creative Cloud proficiency: high

Other Experience

2022 – Present: Website, design, and general IT responsibility for the FACET group, NTNU